



Name: _____

Total Marks: 32 marks

Total Time: 40 minutes

Show working and answers on this sheet. Show working in sufficient detail to support your answers. Incorrect answers without supporting reasoning may be allocated zero marks.

Resources Allowed – 1 A4 Page (1 Side) of Notes, Calculator

Question 1**4 marks**

- a. How much interest is earned on a Principal of \$4000, invested at a rate of 6% per annum for a term of 5 years?

A \$1495 **B** \$1200
 C \$1500 D \$1800

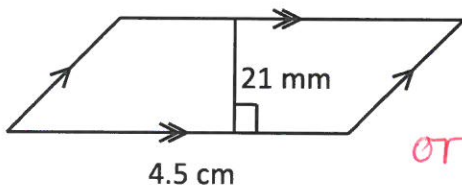
- b. A washing machine costs \$700. If it is paid off over three years at 15% per annum simple interest, how much interest is paid?

A \$105 **B** \$315
 C \$420 D \$805

Question 2**10 marks**

Find the area of the following. Give answer to 1 decimal place.

a.

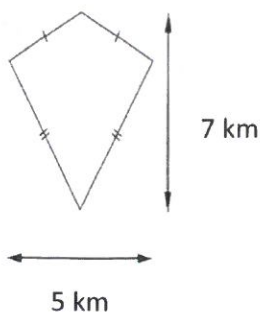


$$4.5 \times 2.1 = 9.5 \text{ cm}^2$$

$$\text{or } 45 \times 21 = 945 \text{ mm}^2$$

[2 Marks]

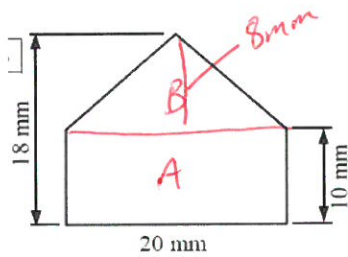
b.



$$\frac{1}{2} \times 7 \times 5 = 17.5 \text{ km}^2$$

[2 Marks]

c.



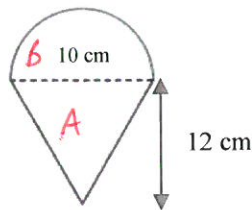
$$A = 10 \times 20 = 200 \text{ mm}^2$$

$$B = \frac{1}{2} \times 8 \times 20 = 80 \text{ mm}^2$$

$$\text{Total} = 280 \text{ mm}^2$$

[3 Marks]

d.



$$A = \frac{1}{2} \times 10 \times 12 = 60 \text{ cm}^2$$

$$B = \frac{1}{2} \times \pi \times 5^2 = 39.3 \text{ cm}^2$$

$$\text{Total} = 99.3 \text{ cm}^2$$

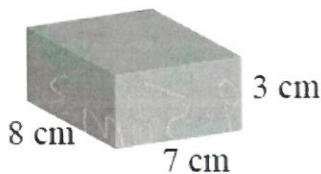
[3 Marks]

Question 3

4 marks

Find the surface area of each of the following prisms:

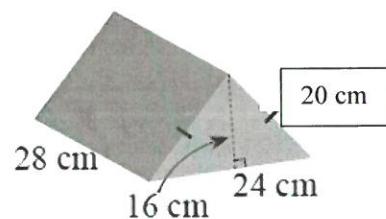
i)



$$(8 \times 7 \times 2) + (7 \times 3 \times 2) + (8 \times 3 \times 2)$$

$$= 202 \text{ cm}^2$$

ii)



$$2 \times \frac{1}{2} \times 24 \times 16 = 384$$

$$2 \times 20 \times 28 = 1120$$

$$24 \times 28 = 672$$

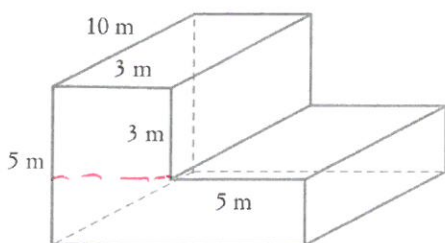
$$\text{Total } 2176 \text{ cm}^2$$

Question 4

3 marks

This diagram represents a new storage building.

Find:



- a the area of the front.

$$3 \times 3 + 2 \times 8 = 25 \text{ m}^2$$

[2 marks]

- b the volume of the building.

$$25 \times 10 = 250 \text{ m}^3$$

[1 mark]

Question 5

5 marks

- a. Mike took out a loan with an interest rate of 12.5% over 4 years. The interest he paid on the loan was \$3250. Calculate the Principal amount he originally borrowed.

$$P = \frac{I}{RT}$$

$$P = \frac{3250}{0.125 \times 4}$$

$$P = \$6500$$

- b. Joe buys a car worth \$24 000. He makes payments of \$550 every month for five years. What rate of interest is he paying?

$$550 \times 12 \times 5 = 33000$$

$$33000 - 24000 = 9000 \text{ Interest.}$$

$$\text{Rate} = 7.5\%$$

$$R = \frac{I}{P \cdot T}$$

$$R = \frac{9000}{24000 \times 5}$$

$$= 0.075$$

Question 6**6 marks**

A room has ceilings 250 cm high and is 5 m by 370 cm. Ignoring doors and windows, find the surface area of the walls and the cost of painting the walls at \$14.50/m².

$$250 \text{ cm} = 2.5 \text{ m.}$$

$$370 \text{ cm} = 3.7 \text{ m.}$$

$$2.5 \times 5 \times 2 + 2.5 \times 3.7 \times 2 = 43.5 \text{ m}^2$$

$$43.5 \times 14.50 = \$630.75$$